Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/25/2005
PI Name:	Smith, Fredrick D B.S.		
Project Title:	Advanced Trace Contaminant Control Systems Development		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	frederick.d.smith1@jsc.nasa.gov	**Fax:**	FY 281 483-5060
PI Organization Type:	NASA CENTER	**Phone:**	281 483-9232
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code EC3		
PI Address 2:	Building 7, Room 222		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	06/01/1999	**End Date:**	09/26/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	none		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Paradoxically, trace contaminant control systems that suffer unexpected upsets and malfunctions can release hazardous gaseous contaminants and cause toxicological problems. This project has characterized hazards and upsets with earlier Trace Contaminant Control System (TCCS) systems, and has focused on a design and prototype construction that addresses these issues. The task objective is to develop a regenerative TCCS that improves on the International Space Station (ISS) baseline TCCS. The task for the FY04 period include: a. Advanced Trace Contaminant Control System control software development. b. Full system testing.		

	c. Preliminary flight hardware design. d. Documentation Package. e. Ground system delivery, installation, and training. Task looks at the enabling question of contaminant control as part of atmosphere management.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved trace gas removal processes could affect a variety of air quality applications, particularly indoor air quality in commercial office space.
Task Progress:	Progress for this reporting period for the following task includes: a. Advanced Trace Contaminant Control System control software development - completed. b. Full system testing - 80 percent complete. c. Preliminary flight hardware design - to be completed in early calendar year 2005. d. Documentation Package- to be completed with delivery and installation of hardware early calendar year 2005. c. Ground system delivery, installation, and training - to be completed early calendar year 2005.
Bibliography Type:	Description: (Last Updated: 02/27/2004)